

Project 1C

Local versus Global Redundancy Planning on KUKA iiwa 7

Motivation

The KUKA iiwa 7 has one more joint than is required for a general six-dimensional Cartesian task. This redundancy can be used to avoid joint limits and singularities, but a local redundancy-resolution law does not know the global structure of the feasible IK families. A locally reasonable motion can therefore drift into a dead end even when another redundancy corridor would complete the task.

The central question of this project is:

When does local null-space redundancy resolution fail, and how can a global IK map reveal a feasible trajectory that the local method does not discover?

Task

You must reconstruct the iiwa 7 kinematics from the supplied URDF, build consistent DH and screw representations, and verify the forward kinematics through both DH and Product of Exponentials. From the screw axes, derive the complete 6×7 Jacobian and verify it numerically.

Use q_3 as a redundancy coordinate. When $q_3 = \lambda$ is fixed, construct the reduced 6×6 Jacobian for the remaining joints and make a clear distinction between loss of rank in this reduced chart and a genuine singularity of the full seven-joint robot. A fixed- q_3 analytical IK routine will be supplied for this project; you must independently verify every returned solution and preserve the supplied branch labels only when the numerical configurations support them.

Choose a full-pose Cartesian path that drives at least one reasonable initial configuration toward a joint limit or a narrow feasibility region. First implement a local redundancy-resolution strategy based on a pseudoinverse plus a null-space objective for joint-limit avoidance. The end-effector task must be corrected throughout so that apparent improvement in joint posture is not obtained by sacrificing Cartesian accuracy.

Next, use the fixed- q_3 IK to construct a global path-progress versus q_3 feasibility map. At every path sample, distinguish absence of real IK, joint-limit violation, poor conditioning of the full 6×7 Jacobian, and singularity of the reduced fixed- q_3 chart. Identify the continuous feasible corridors that remain after these tests.

Use this map to plan a varying- q_3 trajectory through the path, for example with a layered graph or dynamic program, *you can use a naive approach that chooses the nearest valid solution in the next step of the path*. The global and local methods must start from the same initial robot configuration and satisfy the same kinematic constraints. Compare their behavior and determine whether a local failure is caused by numerical integration, by the chosen secondary objective, or by an inability to discover another globally feasible redundancy corridor.

Expected Output

- A validated DH and screw model of the KUKA iiwa 7.
- Numerical agreement between URDF, DH, and PoE forward kinematics.
- The complete 6×7 screw Jacobian and the reduced fixed- q_3 6×6 Jacobian.
- Independent verification of the supplied fixed- q_3 IK solutions and their branch identities.
- A local null-space joint-limit-avoidance controller with Cartesian task correction.
- A path-progress versus q_3 feasibility map separating no-IK, joint-limit, full-Jacobian, and reduced-chart failures.
- Identification of one or more continuous feasible redundancy corridors.
- A globally planned varying- q_3 trajectory starting from the same initial configuration as the local method.
- A quantitative comparison of local and global methods using completion, Cartesian error, joint travel, minimum joint-limit margin, and full-Jacobian singularity margin.
- A diagnosis of whether any local failure is numerical, objective-induced, or genuinely caused by loss of feasible connectivity.

Useful resources

1. Simulations to visualize your robots: <https://epfl-lasa.github.io/eng-654/#simulations>
2. Robot assets (URDF and .stl files for visualization): <https://epfl-lasa.github.io/eng-654/#download-assets>